Two-Sample t-Test

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Introduction

The two-sample t-test is the default procedure for deciding whether the means of two independent groups differ. In its modern form it comes in two flavours: Student's t-test, which assumes equal variances in the two populations, and Welch's t-test, which does not. Both produce a test statistic, a p-value, and a confidence interval for the difference in means.

Prerequisites

A reader should understand the difference between a population mean and a sample mean, the concept of a sampling distribution, and the logic of hypothesis testing. Familiarity with R's formula interface is assumed.

Theory

Let $X_1, \ldots, X_{n_1}$ be independent observations from $\mathcal{N}(\mu_1, \sigma_1^2)$ and $Y_1, \ldots, Y_{n_2}$ independent observations from $\mathcal{N}(\mu_2, \sigma_2^2)$. We test $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 \neq \mu_2$.

Student's test assumes $\sigma_1^2 = \sigma_2^2 = \sigma^2$ and uses the pooled variance estimator. The test statistic

$$T = \frac{\bar{X} - \bar{Y}}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

follows a t distribution with $n_1 + n_2 - 2$ degrees of freedom under H_0, where s_p^2 is the pooled sample variance.

Welch's test allows $\sigma_1^2 \neq \sigma_2^2$ and uses the Satterthwaite approximation:

$$T = \frac{\bar{X} - \bar{Y}}{\sqrt{s_1^2/n_1 + s_2^2/n_2}},$$

with degrees of freedom

$$\nu = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}.$$

Modern recommendations favour Welch's test as the default, because the assumption of equal variances is rarely verifiable in practice and the loss of efficiency when the variances *are* equal is small.

Assumptions

Both tests assume:

1. Observations within each group are independent of each other.
2. The two groups are independent of each other.
3. The observations in each group are approximately normal, or the sample size is large enough for the CLT to apply to the sample means.

Student's additionally assumes equal variances. Welch's does not.

R Implementation

```
library(effectsize)

set.seed(42)
group_a <- rnorm(30, mean = 100, sd = 15)
group_b <- rnorm(35, mean = 110, sd = 18)

df <- data.frame(
  score = c(group_a, group_b),
  group = factor(rep(c("A", "B"), c(30, 35)))
)

t.test(score ~ group, data = df)

cohens_d(score ~ group, data = df)
```

The default `t.test()` in R performs Welch's test. For Student's, set `var.equal = TRUE`. The `effectsize::cohens_d()` function returns Cohen's d with a confidence interval.

Output & Results

Typical output:

```
        Welch Two Sample t-test

data:  score by group
t = -2.3415, df = 61.832, p-value = 0.02241
alternative hypothesis: true difference in means between group A and group B is not equal to 0
95 percent confidence interval:
 -18.8912  -1.4872
sample estimates:
mean in group A mean in group B
       100.4510        110.6407
```

Cohen's d is approximately -0.59 with 95% CI $(-1.09, -0.09)$ – a moderate effect.

Interpretation

The conventional report is: "group B had a mean score of 110.6 (SD 18.2), which was significantly higher than group A's mean of 100.5 (SD 14.5); Welch's $t(61.8) = -2.34$, $p = 0.022$, mean difference = 10.2 (95% CI 1.5 to 18.9), Cohen's $d = -0.59$." The effect-size reporting is essential: a p-value alone does not tell the reader how large the difference is.

Practical Tips

- Always report the effect size and its confidence interval alongside the p-value; journals increasingly require this.
- Prefer Welch's test as the default; only use Student's when equal variances are genuinely plausible (for example, a randomised experiment with balanced groups).
- Check normality with Q-Q plots rather than with Shapiro-Wilk – the latter has little power at small n and rejects trivially at large n.

- For strongly skewed or outlier-contaminated data, consider the Wilcoxon rank-sum test, a permutation test, or analysis on a log scale.
- Unequal sample sizes are fine; extreme imbalance (say, 10 vs. 1000) warrants extra care because variance assumptions and power are both affected.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests